What Conformal Gravity Must Assume
to Explain Galactic Rotation Curves

Claude Opus 5* Asger Alstrup Palm†

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Abstract

Conformal (Weyl) gravity admits a static spherically symmetric vacuum solution whose metric function carries a term linear in r. That term is the basis of a long-standing proposal to fit galactic rotation curves without dark matter. We ask a reverse-physics question about it: not whether the fits succeed, but *what must be assumed for them to mean anything.*

Seven results, each an exact symbolic computation with machine-checked controls. (i) The linear term is not an ansatz: in the conformal gauge the Bach vacuum equation is exactly the *biharmonic* equation $\nabla^4 B = 0$, so γr is the point-source response of the governing operator, on the same footing as the Newtonian $1/r$. (ii) If the baryonic Tully–Fisher relation holds, γ is forced to be a universal constant and is identified with the MOND acceleration scale, $\gamma = a_0/2c^2$; a γ carrying a mass-proportional piece gives the wrong slope. (iii) Conformal invariance forces the source to be trace-free, so a perfect fluid must satisfy $p = \rho/3$ and pressureless baryons are excluded outright. (iv) The standard repair—breaking the symmetry with a conformally coupled scalar—fails in its simplest form: a *constant* vacuum expectation value pins the Ricci scalar and thereby forces $\gamma = 0$, removing the very term at issue.

Two further results say what the surviving branch costs. (v) γ is not a conformal invariant: the conformal group acts on the four parameters of the general solution through a single Möbius map of the radius, and shifts γ. The frame in which the scalar is constant is reached by exactly that map, so for a non-constant scalar the linear term is generated by the profile, with $\gamma = -2(S'/S) - 3u(S'/S)^2$. The demand that γ be universal is therefore the demand that the scalar's *fractional radial gradient* be universal. (vi) Which frame matter follows is a variational identity, not an interpretation: a mass generated by the scalar satisfies $-\int hS\,\mathrm{d}s_g = -\int h\,\mathrm{d}s_{S^2 g}$, so the mass scale *is* the conformal factor and such a particle follows geodesics of the frame where $\gamma = 0$. A particle whose mass does not track the scalar follows the other frame, where γ acts.

(vii) That pair is an exact solution and not a relabelling—in the constant-scalar frame the Einstein condition and Bach flatness hold together, with the de Sitter term equal to the scalar potential—and u is a conformal invariant while γ is not. Two frames with the same mass carry different γ, so γ is *not a function of the mass.*

The conclusion is therefore sharper than a coincidence in need of explanation. Universality of γ is not in tension with the structure, because γ cannot depend on mass at all; it is the statement that one boundary datum on the symmetry-breaking sector is common to every galaxy while the mass varies freely. That is consistent, it is not delivered by the gravitational theory, and it is the whole of what a conformal-gravity rotation-curve fit assumes. We do not adjudicate it. We locate it, and the location is one number.

*Anthropic model; computations and drafting.

†Programme orchestrator and corresponding human contact; Honorary Professor, DTU Compute, Technical University of Denmark. `asger@area9.dk`.

1 The question

Conformal gravity has been proposed as an explanation of galactic rotation curves without dark matter, on the strength of a term linear in r in its static spherically symmetric vacuum solution. The literature on whether the fits succeed is substantial and we add nothing to it.

We ask instead the reverse-physics question: *what has to be true for the explanation to be an explanation?* A fit with a per-galaxy free parameter is a description; a fit with a parameter forced by the theory is a prediction. Which of these the proposal is depends on facts that can be settled by computation, and this paper settles four of them.

Every statement below is an exact symbolic identity over $\mathbb{Q}$, carried out with unspecified functions rather than sampled fixtures, with controls designed to fail if the claim were vacuous. Certificates and verifier commands are listed in §9.

Conventions. We work in $D = 4$, signature $(-, +, +, +)$, with the static spherically symmetric ansatz

$$\mathrm{d}s^2 \;=\; -B(r)\,\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{B(r)} + r^2\mathrm{d}\Omega^2,$$

the conformal gauge $g_{tt}g_{rr} = -1$. §6 addresses what that gauge costs. B_{ab} denotes the Bach tensor.

2 The linear term is forced, and it is a point-source response

Theorem 2.1 (Completeness without an ansatz). *Let B be any C^4 function, nonvanishing on an interval. Then $B_{ab} = 0$ identically if and only if*

$$(rB)'''' = 0 \qquad \textit{and} \qquad c_1^2 - 3c_0c_2 = 1,$$

where $B = c_0/r + c_1 + c_2 r + c_3 r^2$.

The proof is a decomposition. With $L := (rB)''''$ and N the nonlinear remainder, the three independent Bach rows satisfy, as exact identities in B and its derivatives,

$$B_{\theta\theta} = \frac{N}{24r^2}, \qquad B_{tt} = \frac{B\,(N + 2r^3BL)}{24r^4}, \qquad B_{rr} = \frac{-N + 2r^3BL}{24r^4B},$$

so all three lie in the span of $\{N,\ r^3BL\}$, and both generators are recoverable from the rows. Hence $B_{ab} = 0$ forces $L = 0$ and $N = 0$.

The essential point is that L is *linear*. Substituting $u = rB$ turns $L = 0$ into $u'''' = 0$, whose solution space is the cubics by four integrations—no ansatz. On that space N collapses to the constant $4(1 + 3c_0c_2 - c_1^2)$, leaving one algebraic condition.

Theorem 2.2 (The vacuum equation is biharmonic). *In spherical symmetry $\nabla^2 f = f'' + 2f'/r$, and for unspecified B*

$$\nabla^4 B \;=\; B'''' + \frac{4B'''}{r} \;=\; \frac{1}{r}\,(rB)''''.$$

So Theorem 2.1's linear equation is $\nabla^4 B = 0$: the Bach vacuum condition says B is *biharmonic*. The forced solution space $\{1/r,\ 1,\ r,\ r^2\}$ is the radial biharmonic kernel, and r^3 and r^{-2}—its nearest neighbours—are verified not to lie in it.

Corollary 2.3 (γr is a point-source response). *For ∇^2 the point-source potential is $1/r$ and its coefficient is the mass; this is Newton, and the origin of $\beta = GM/c^2$. For ∇^4 the corresponding object is r, since $\nabla^2 r = 2/r$ is harmonic away from the origin.*

This is a stronger statement than "γr survives the equation". It is the term the equation *produces* for a localised source, and it sits on the same footing as the Newtonian term rather than being appended to it.

Remark 2.4. That the general static spherically symmetric vacuum solution of conformal gravity has this form is classical. What is contributed here is the removal of the ansatz and the identification of the operator, both computed for an unspecified function.

3 Tully–Fisher forces γ to be universal

With $\beta = GM/c^2$ the locally measured circular velocity is $v^2/c^2 = rB'/2B$, which at small field strength is $\beta/r + \gamma r/2 - kr^2$. In the galactic regime this has a stationary point, and it is a minimum:

$$r_* = \sqrt{2\beta/\gamma}, \qquad v^4 = 2GM\gamma c^2\Big(1 - \tfrac{3}{2}\beta\gamma\Big).$$

Theorem 3.1 (The conditional)**.** *To leading order in $\beta\gamma$: if γ is universal then $v^4 \propto M$, the baryonic Tully–Fisher relation with slope 4 in v. If γ carries a piece proportional to M then $v^4 \propto M^2$. Matching the observed $v^4 = GMa_0$ forces*

$$\gamma \;=\; \frac{a_0}{2c^2}.$$

Read in the reverse-physics direction: the observed Tully–Fisher relation, combined with the forced metric, requires γ to be a universal constant of nature rather than a per-galaxy parameter, and identifies it with the MOND acceleration scale. That is a prediction rather than a fit, and it is falsifiable in a specific way—a mass-dependent γ moves the slope.

Two boundaries travel with it. Identifying $v_{\min}$ with the observed flat velocity is an interpretive step, not a theorem. And the correction factor $1 - \frac{3}{2}\beta\gamma$ is itself mass-dependent, so the slope is exactly 4 only in the limit $\beta\gamma \to 0$; for a galaxy the departure is of order 10^{-14} and unobservable, but "exactly linear" would be too strong.

Remark 3.2. Conformal gravity and MOND are not the same theory and disagree away from r_*: beyond it the conformal curve *rises* ($\mathrm{d}v^2/\mathrm{d}r|_{2r_*} = 3\gamma/8 > 0$) where MOND's are asymptotically flat. Only the $-kr^2$ term can bend it back. The outer rotation curve is therefore a discriminant, not a detail.

4 What may source the theory

§2 and §3 concern the vacuum. γ is a free coefficient there; the explanatory claim needs it *sourced*.

A conformally invariant action satisfies $g_{ab}\,\delta S/\delta g_{ab} = 0$. Applied to the gravitational action this is the Noether identity of the theory; applied to the matter action it reads $T^\mu{}_\mu = 0$. It is one theorem used twice, and we take it as given.

Theorem 4.1 (Trace obstruction)**.** *A perfect fluid has $T^\mu{}_\mu = 3p - \rho$, so it is trace-free if and only if $p = \rho/3$. Dust, $p = 0$, forces $\rho = 0$.*

Radiation, uniquely, and pressureless matter not at all. A galaxy's baryons are pressureless to excellent approximation, so on the conformally invariant branch a galaxy is not an admissible source.

The standard repair is to break the symmetry with a conformally coupled scalar acquiring a vacuum expectation value, generating a scale so that ordinary matter can source the theory after all. In its simplest form this fails, and it fails on the term at issue.

Theorem 4.2 (A constant vacuum expectation value forces $\gamma = 0$)**.** *For the conformally coupled scalar,* $\Box S + \frac{1}{6}RS + 4\lambda S^3 = 0$ *reduces for constant* $S = S_0 \neq 0$ *to* $R = -24\lambda S_0^2$*, a constant. For the forced family,*

$$R \;=\; 12k - \frac{6\gamma}{r} + \frac{2(1-w)}{r^2},$$

which is constant in r *if and only if* $\gamma = 0$ *and* $w = 1$*.*

A constant vacuum expectation value does not merely supply a scale; it pins the Ricci scalar, and the forced family is compatible with a pinned R only where the linear term vanishes. Note that u, the Newtonian coefficient, drops out of R entirely: the obstruction is specific to γ.

5 The trichotomy

Collecting §4, the possibilities are organised by one question—*is there a scale, and is it constant or field-dependent?*

	scale	consequence
(A) conformal matter	none	source must be trace-free, hence radiation; a galaxy is excluded, so γ is not tied to M and Tully–Fisher is a coincidence
(B1) constant $\langle S\rangle$	constant	R pinned, hence $\gamma = 0$; the linear term is removed
(B2) non-constant S	field-dependent	γ tied to the scalar profile; §7 makes the tie explicit

(A) does not remove γ; it decouples γ from mass, which destroys the explanation while leaving the term. (B1) removes γ outright. Exactly one branch survives, and it is the one the literature in fact uses.

Corollary 5.1 (Where the assumption lives)**.** *On branch (B2),* γ *is determined by the scalar profile, which is a property of each configuration. Theorem 3.1 requires* γ *to be the same constant for every galaxy. Therefore the requirement transfers to the profile.*

That is a question about the symmetry-breaking sector rather than about gravity. §7 carries out the transfer, and it turns out to land on one derived quantity rather than on the profile as a whole.

6 What the conformal gauge costs

Sections 2–5 work in the gauge $g_{tt}g_{rr} = -1$. Two things are needed to justify it.

Proposition 6.1 (Conformal covariance)**.** *In* $D = 4$*,* $B_{ab}[\Omega^2 g] = \Omega^{-2}B_{ab}[g]$*, verified for the Mannheim–Kazanas family with symbolic parameters and* unspecified Ω*. Hence Bach flatness is a property of the conformal class, and the classification of §2 classifies classes.*

Proposition 6.2 (Reachability is a first-order ODE)**.** *Setting the new radius to* $\rho = \Omega r$ *so the angular part remains* ρ^2*, the gauge condition becomes*

$$r\,\Omega' \;=\; \Omega\big(\Omega\sqrt{ab} - 1\big),$$

first order in Ω*, hence locally solvable wherever* $ab > 0$ *and* $r \neq 0$*.*

What remains assumed is only global continuation—whether Ω stays positive and finite across the exterior. That is strictly smaller than the original assumption and is a question of analysis rather than algebra.

7 Carrying out the transfer

§6 and §2 between them force a third statement that neither makes on its own. Bach flatness is a property of a conformal class; the general solution in the gauge $g_{tt}g_{rr} = -1$ is four parameters subject to one constraint. So the conformal group must *act* on those four parameters.

Theorem 7.1 (The conformal action on the forced family). *Staying in the gauge is not a choice but an ODE,* $\mathrm{d}\hat{r}/\mathrm{d}r = (\hat{r}/r)^2$*, whose general solution is the Möbius map* $\hat{r} = r/(1 - Cr)$*—one constant. The induced action is*

$$u \mapsto u, \qquad w \mapsto w + 3uC, \qquad \gamma \mapsto \gamma - 2wC - 3uC^2, \qquad k \mapsto k + C\gamma - C^2 w - C^3 u,$$

a one-parameter group preserving $w^2 + 3u\gamma$ *identically. From a member with* $\gamma = 0$ *it produces* $\gamma = -2C - 3uC^2$.

So γ is not an invariant of the Bach vacuum. It is a coordinate on the conformal class, and the classification of §2 classifies classes precisely because of this.

A conformally coupled scalar has weight -1, so the frame in which it is *constant* is reached by exactly this map with $\Omega = S/S_0$—and Theorem 4.2 puts $\gamma = 0$ there. The transfer follows.

Corollary 7.2 (The transfer, in closed form). *On branch (B2) the profile realising the frame change is* $S(r) = S_0/(1 - Cr)$ *with* $C = (S'/S)$ *at the origin, and*

$$\gamma \;=\; -2\,(S'/S) \;-\; 3u\,(S'/S)^2 \qquad \text{so that} \qquad \gamma \simeq -2\,(S'/S).$$

Hence Theorem 3.1's demand that γ *be one constant for every galaxy is exactly the demand that* S'/S *be one constant for every galaxy.*

That is more specific than "is the profile universal?". Only one derived quantity has to be universal, and it has to be so across galaxies whose masses differ by orders of magnitude, for a field sourced by the configuration.

Which frame matter follows

Both frames describe the same Bach vacuum, and only one carries a linear term. So the disagreement of [3, 4] is not about the vacuum solution and not about the algebra: it is about which frame matter follows. That premise is itself computable.

Theorem 7.3 (The mass scale is the conformal factor). *A point particle of mass* m *extremises* $-\int m\,\mathrm{d}s$*. If the mass is generated by the symmetry-breaking scalar,* $m = hS$*, then*

$$-\int hS\,\mathrm{d}s_g \;=\; -\int h\sqrt{-S^2 g_{ab}\,\mathrm{d}x^a \mathrm{d}x^b} \;=\; -\int h\,\mathrm{d}s_{S^2 g}$$

identically, so the trajectory is a geodesic of $S^2 g$ *and not of* g*. Since* $S^2 g$ *is the frame in which* S *is constant up to the constant* S_0^2*, such a particle follows geodesics of the frame with* $\gamma = 0$.

Both branches are exhibited with their circular-orbit velocities, and both frames are confirmed Bach vacua, so this is a fork between two admissible solutions of one theory:

mass	trajectory follows	linear coefficient of v^2
m constant, independent of S	geodesics of g	$\gamma/2$
$m = hS$, generated by S	geodesics of $S^2 g$	0

The first entry reproduces Theorem 3.1's weak-field formula by a different route. The disagreement is not an artefact of the expansion: the exact circular-orbit velocities differ too.

A rotation-curve fit needs the first row. The second row is what the symmetry breaking was introduced to provide. Whether a matter sector can generate a mass scale by breaking conformal invariance while leaving particle masses independent of the breaking field is a question about that sector, and we do not answer it. What is no longer available is leaving the choice implicit.

The pair is a solution, and γ is independent of the mass

Theorem 7.1 produces a profile from a transformation law, and a transformation law does not by itself say the result solves anything.

Theorem 7.4 (Admissibility)**.** *In the constant-scalar frame the conformally coupled action collapses to Einstein–Hilbert plus a cosmological term, so its stress tensor is a combination of G_{ab} and g_{ab}. The gravitational equation is $\alpha B_{ab} = \frac{1}{2}T_{ab}$ and that frame's metric is a Bach vacuum, so consistency forces $T_{ab} = 0$—for a constant scalar, the Einstein condition. The metric $1 - u/r - \hat{k}r^2$ is Einstein, with $R = 12\hat{k}$; equating with Theorem 4.2's $R = -24\lambda S_0^2$ gives $\hat{k} = -2\lambda S_0^2$. Bach flatness and the Einstein condition then hold together on the same metric.*

The de Sitter term *is* the scalar potential, and the exterior pair solves the coupled system with nothing left over.

Corollary 7.5 (γ is not a function of the mass)**.** *Theorem 7.1 gives $u \mapsto u$, so the Newtonian coefficient is a conformal invariant of the solution while γ is a coordinate on the class. Two frames with the same u carry different γ. Hence γ requires a second, independent datum—namely C—and for any target γ and any u there is a C attaining it.*

This changes the standing of Theorem 3.1. Requiring γ to be one constant across galaxies of widely differing mass looked like a coincidence in need of explanation; it is not one, because γ cannot depend on mass in any case. What the requirement says is that a single boundary datum on the symmetry-breaking sector is common to every galaxy.

Both readings of that are available and we prefer neither. Favourably, the universality is structurally natural rather than a fine-tuning of gravity. Unfavourably, the explanation of rotation curves is imported from the symmetry-breaking sector rather than produced by gravity.

8 What this paper does not establish

- **No novelty in the physics.** The Mannheim–Kazanas solution, its reduction to a fourth-order Poisson problem, and the trace-free condition on conformal matter are all classical. The contribution is that each is computed here with the ansatz removed and the assumptions isolated, rather than cited.

- **No fits.** No galaxy data enters anywhere. The only observational inputs are the *shape* of the Tully–Fisher relation, used as a premise in Theorem 3.1, and that galaxies are pressureless, used in §4. Neither is our measurement.
- **No interior.** Theorem 7.4 closes the *exterior*: the pair is an exact solution of the coupled system in vacuum. What is not done is the interior, and the matching that would fix C from a given configuration; that needs a matter model we do not have, and it is what remains of "forced versus sourced". Nor is it shown that a cosmological boundary condition *can* deliver a universal C. The assumption is named, not discharged.
- **No resolution of the Flanagan–Mannheim dispute** [3, 4] over the Newtonian limit under conformal matter coupling. Theorem 7.3 narrows it to one premise and computes both branches, so each side's commitment is visible; it does not say which a real matter sector realises. The Yukawa form $m = hS$ is itself a premise, standard but entered rather than derived.
- **Nothing about the ghost.** Conformal gravity's fourth-order kinetic operator carries an Ostrogradsky ghost. That is orthogonal to everything here and is treated elsewhere in this programme.
- **No numerical value.** a_0 is not measured here and no published γ is compared; $\gamma = a_0/2c^2$ is an algebraic consequence of a premise, not a numerical agreement.

9 Verification

Each result is a fail-closed certificate with a deterministic verifier. Controls are designed to fail if the claim were vacuous; several did so during development and are recorded in the certificates.

result	certificate	checks
Thm. 2.1	`BHOB_GENERAL_STATIC_SPHERICAL_COMPLETENESS`	24
Thm. 3.1	`BHOC_TULLY_FISHER_SCALING`	14
Thm. 2.2	`BHOD_BACH_VACUUM_IS_BIHARMONIC`	9
Thm. 4.1	`BHOE_MATTER_TRACE_DILEMMA`	8
Thm. 4.2	`BHOF_CONSTANT_VEV_FORCES_GAMMA_ZERO`	12
§6	`BHOG_CONFORMAL_CLASS_INVARIANCE`	10
Thm. 7.1	`BHOH_UNIVERSALITY_TRANSFER`	18
Thm. 7.3	`BHOI_WHICH_FRAME_MATTER_FOLLOWS`	13
Thm. 7.4	`BHOJ_THE_ASSUMPTION_NAMED`	12

Theorems 2.1, 4.2, 7.1, 7.3 and 7.4, and §6, are computed in SYMPY; Theorems 2.2 and 4.1 are computed in Forge over exact rationals with symbolic differentiation of unspecified functions. Theorem 2.1's conclusion carries an independent Forge rail using a different arithmetic stack, representation and code path; that rail confirms which metrics are Bach-flat but not the row decomposition itself, which remains single-implementation.

References

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